MST REVIEW

SURFACE TEXTURE

surface lay symbols

symbol	type
=	parallel lay
\perp	perpendicular lay
X	crossed/oblique lay
M	multidirectional
C	circular
R	radial
P	particulate

LIMITS AND FITS

GD&T

Table 1.27 Symbols for geometric characteristics

Geometric characteristic	*Symbol*	*Type of tolerance*	*Application*
Straightness	—	Form	For individual features
Flatness	▱		
Circularity	○		
Cylindricity	⌭		
Profile of a line	⌒	Profile	For individual or related features
Profile of a surface	⌓		
Angularity	∠	Orientation	For related features
Perpendicularity	⊥		
Parallelism	//		
Position including concentricity and symmetry	⌖	Location	
Circular runout	↗	Runout	
Total runout	⌰		

$$\text{Effective Tolerance} = |\text{MMC Size} - \text{Actual Size}| + \text{Stated Tolerance}$$

$$\begin{aligned}\text{Ring Gauge} &= \text{MMC Size} + \text{MMC Tolerance} \\ &= \text{Actual Size} + \text{Effective Tolerance}\end{aligned}$$

$$\begin{aligned}\text{Pin Gauge} &= \text{MMC Size} - \text{MMC Tolerance} \\ &= \text{Actual Size} - \text{Effective Tolerance}\end{aligned}$$

DFMA

- REDUCE PART COUNT
  - Reduce Part Count
  - Reduce Screw/Screw Type Count
  - Make Parts Multifunctional
  - Allow Modular Assemblies
  - **Allow Part Interchangability
- MAKE ASSEMBLY EASIER
  - Facilitate Part Handelling
  - Allow Open Assemblies
  - Dont Fight Gravity
  - Minimise Surface Count
  - Design Self Locators
  - Eliminate Part Interfaces
- OPTIMISE MANUFACTURING PROCESSES
  - Optimise Manfuacturing Process
  - Use Standard Parts
  - Maximise Design Tolerances

DFMG

- DRILLING:
  - INHERENT PROCESS PROPERTIES
    - Justify Drilling Need
    - Cast Large Holes
    - Avoid Boring Holes
  - GOOD DESIGN CHOICES
    - Prefer Through Holes
    - Avoid Thin Holes
    - Avoid Flat Bottoms
    - Minimise Hole Types
    - Use Standard Sizes
    - Dimension to Rectangular Datums
  - OPTIMISING MANUFACTURING PROCESSES
    - Avoid Hole Overlap
    - Give Sufficient Clearance
    - Space Out Holes
    - Drill Perpendicular Surfaces
    - Sufficiently Drill Material
- MILLING:
  - INHERENT PROCESS PROPERTIES
    - Avoid Deep Corner Radius
    - Avoid Sharp Internal Corners
    - Avoid Undercuts
    - Avoid Deep Narrow Slots
  - GOOD DESIGN CHOICES
    - Use Standard Tool Sizes
    - Prefer Chamfers to Filets
    - Use Bosses for Flatness
    - Avoid Thin Features

- Allow Internal Radii Leeway
- Allow Uniform Draft Features
- OPTIMISING MANUFACTURING PROCESSES
  - Allow Cutting Tool Accessibility
  - Use Thicker Cutting Tools
- TURNING:
  - INHERENT PROCESS PROPERTIES
    - Avoid Sharp Corners
    - Avoid Slender Parts
    - Minimise Excessive Machining
  - GOOD DESIGN CHOICES
    - Avoid Tight Tolerances
    - Specify Burr Removal
    - Minimise Perpendicular Steps
    - Adapt to Function
  - OPTIMISING MANUFACTURING PROCESSES
    - Allow Tool Runout
    - Avoid Clamping on Partings
    - Turning in Avoidance Zones
    - Make Surfaces Accessible
    - Allow Sufficient Clamping
    - Allow Cast-In Relief

DFJA

- WELDMENTS:
  - GOOD DESIGN CHOICES
    - Weld Similar Thicknesses
    - Weldment Supporting Loads
    - Specify Weldment Order
  - OPTIMISE MANUFACTURING PROCESSES

Avoid Final Welding

Provide Sufficient Access

Avoid Rigid Welding

Avoid Machining Weldments

THREADS

Design for Manufacture and Assembly (DFMA)

1. List the four primary benefits achieved when applying Design for Manufacture and Assembly (DFMA) principles to product development.
    a. Reduced Production Cost, Quicker Assembly Times, Reduced Material Cost, XXXXXX
2. Design for Manufacturing (DFM) includes considerations for materials, fabricated parts, and product assembly. Provide one specific question a designer should consider for each of these three areas.
    a. What materials are required for the part's function?
    b. What fabrication process can reduce the cost of manufacture?
    c. How can the part be designed to optimise assembly times?
3. A core principle of DFMA is to minimize the number of parts. List the three critical questions that can be asked to determine if a component can be eliminated or integrated with another part.
    a. Does the part need to be a different material?
    b. Does the part need to move relative to other parts?
    c. Can the part be combined with other parts?
4. Describe three different design strategies that ensure parts possess self-locating or self-aligning features, thereby speeding up the assembly process.
    a. chamfers prevent parts from being caught on sharp edges and locate mating parts
    b. tapers/draft angles give more space to allow parts to self locate
    c. spot faced holes self locate bolts into their correct locations

5. Explain the importance of designing for "stack assemblies" (or avoiding fighting gravity) during the assembly operation, particularly in the context of automated assembly.
    a. stack assemblies take advantage of gravity to allow for easier assembly, reducing work needed to align or mate parts as compared to non-stack assemblies
6. When designing tolerances, why is the specification of unnecessarily tight tolerances considered a significant waste of time and money, and what must designers consider when specifying tolerances?.
    a. tighter tolerances require more precise manufacturing, require more expensive tooling, machinery, and slower manufacture times. Designers should know the range of tolerances that the manufacturing process can achieve and design their part accordingly.

Design for Machining (DFM)

1. Why is machining often considered a more expensive manufacturing process for high volumes compared to methods like molding or casting, and what specific design requirements necessitate the use of machining?.
    a. machining requires expensive machinery and tooling and longer setup time, thus being used for low scale precision parts.
2. When designing cylindrical turned parts, what geometric guideline should be followed regarding the aspect ratio to avoid the necessity of using a tail stock?.
    a. the length to diameter ratio should not exceed 3:1.
3. Explain the design guideline regarding internal corners of milled pockets in relation to the cutting tool used. What alternative feature can be used if a sharp internal corner is functionally required?.
    a. internal corners cannot be made excessively sharp, and are limited by the diameter of the tool used, where a sharper corner requires a smaller diameter cutter, which is unfeasible as it more likely to break. An alternative is by drilling a hole at the corner to act as a corner relief, providing two walls that meet at a sharp corner.
4. When designing features for drilling, why are through holes generally preferred over blind holes? If a blind hole must be used, what extra measure should be specified regarding its depth?

   a. through holes allow easier chip removal and coolant flow. For blind holes, the depth must not be 3 times the diameter, otherwise the drill bit may break.

5. **(Application Question)** A part designer specified that all tapped and drilled holes must be positioned on an inclined plane. Explain why this is considered a poor design practice for machining.
   a. drilling on non perpendicular surfaces causes the drill to wander from the hole axis, and can be prevented by making the surface perpendicular by facing to form a horizontal section

Design for Joints and Assembly

1. Define the main difference between "Joining" and "Assembly" processes. Provide one example of a process for each category.
   a. joining refers a permanent bond between parts, like welding, brazing and soldering while assemblies refer to non-permanent bonds that can be disassembled, like rivets, and threaded parts
2. Based on the mechanical method of joining, list three distinct differences between a riveted joint and a bolted joint.
   a. riveted joints are semi-permanent, they can be removed but by being drilled out.
   b. riveted joints only require one side of assembly, unlike bolted joints.
   c. riveted joints are easier and faster to assemble compared to bolted joints.
3. In fusion welding, what design precaution should be taken when machining is required on parts that have been welded, and why is this precaution necessary?
   a. the weldment should not be on the same surface as the machines surface, as somethign soemthing hjbbdhbhdbhjfdjhbbshbjfd
4. Describe the ideal physical orientation for welds to minimize potential stress concentrations.
   a.
5. For standard ISO metric threads, the Medium Fit class is designated as 6H / 6g. What are the characteristics and typical application for this class of fit?.

6. When designing mechanical joints using threaded fasteners, what are the suggested engagement lengths recommended for general clamping, and for applications involving shock and vibration?.